

01 — Scope & Vision

1. Product summary

Solo is a **direct-to-consumer e-commerce platform** for the MENA market (primary: UAE) selling premium lifestyle, travel and home goods. It is a **headless** commerce stack: a NestJS REST API backed by PostgreSQL and a React (Vite) single-page storefront, deployed on Azure.

The platform supports the full retail lifecycle:

- Product discovery (search, browse, filter, variants, recommendations)
- Customer accounts (register, login, addresses, favorites, loyalty wallet)
- Cart and checkout (cards via Stripe, BNPL via Tabby & Tamara, COD)
- Order lifecycle (placement → fulfilment → shipped → delivered → returns)
- Content management (banners, home page sections, navigation, landing pages)
- Admin operations (catalog CRUD, orders, customers, returns, reports)

2. Vision

Be the most trusted premium e-commerce destination in the GCC, with a bilingual (EN/AR), VAT-compliant, mobile-first experience that scales from first sale to enterprise B2B.

3. Business goals

#	Goal	Success measure
G1	Convert visitors to buyers	Conversion rate $\geq$ 1.5% on product pages
G2	Maximise basket size	Average order value (AOV) trending up MoM
G3	Reduce checkout friction	Stripe + Tabby + Tamara + COD all live; $\leq$ 4 steps
G4	Earn repeat purchase	Loyalty wallet AED rebates; favorites; saved cards
G5	Stay VAT-compliant	Per-order VAT snapshot; downloadable PDF invoices
G6	Scale via B2B channel	Bulk-order RFQ flow with admin workflow
G7	Enable rapid merchandising	CMS-driven home, category landing, navigation

4. Personas

- **Shopper (Sara)** — UAE resident, mobile-first, expects Arabic + English, wants Tabby/Tamara split-pay and free returns.
- **Repeat customer (Khalid)** — uses saved cards, follows favorites, redeems loyalty AED at checkout.
- **B2B buyer (Procurement)** — submits a bulk-order RFQ, expects an admin to reply with a quote.
- **Merchandiser (Admin)** — schedules banners, builds home page sections, curates collections, sets sale prices.
- **Operations admin (Super-admin)** — manages stock, orders, returns, customers, reports, settings (VAT rate, shipping tiers).

5. In scope

- B2C storefront (web responsive)
- Catalog with variants (color, size, material), brand & designer associations
- Multi-image, dimension, packaging, specifications, soft-delete tombstones
- Cart with promo codes and loyalty redemption
- Checkout with three providers + COD
- Order management & status history
- Returns with refund methods
- Loyalty wallet (earn / redeem / adjust / expire)
- Admin dashboard (catalog, orders, customers, returns, content, settings)
- Bilingual content (EN + AR fields) with English UI
- VAT-correct pricing (per-line + per-order snapshot, 5% UAE VAT)
- PDF invoices, stock movements ledger
- Bulk-order request capture for B2B leads

6. Out of scope (current release)

- Native mobile apps (web is responsive-first)
- Multi-currency display (AED only — single-currency snapshot fields exist for forward compatibility)
- Multi-warehouse inventory (single virtual warehouse)
- True multi-tenant (single tenant per deployment)
- Marketplace / 3rd-party seller onboarding
- Subscriptions / recurring billing
- In-store POS integration

7. Constraints

- **Region:** Primary infra in Azure `eastus2` ; CDN for region edge caching is served by SWA. PostgreSQL flexible server is single-zone.

- **Compliance:** UAE VAT (5%); future KSA expansion will require 15% rate override (already supported via per-order `vatRateSnapshot`).
- **Identity:** First-party JWT auth (no social/SSO yet).
- **Currency:** AED only at the UI layer; DB carries `currencyCode` columns ready for future multi-currency.
- **Throttling:** `/api/auth/*` is hard-limited to 5 requests / 15 min / IP to resist credential stuffing.

8. Quality attributes (NFRs)

Attribute	Target
Availability	99.5% backend, 99.9% storefront (CDN-served)
API latency p95 (read)	< 700 ms region-internal
Cold start (container)	< 5 s with min replicas = 1
Security	OWASP Top 10 covered (see 08-security)
Observability	Application Insights traces + Pino structured logs
Recoverability	Daily PG flexible-server automated backups
Localization	EN + AR data fields; UI strings English (AR planned)
Accessibility	Keyboard navigable; semantic HTML; alt text required on media

9. Glossary

Term	Meaning
SWA	Azure Static Web Apps (frontend host)
ACA	Azure Container Apps (backend host)
ACR	Azure Container Registry
RBAC	Role-based access control: <code>CUSTOMER</code> , <code>ADMIN</code> , <code>SUPER_ADMIN</code>
BNPL	Buy-now-pay-later (Tabby, Tamara)
Variant	A <code>Product</code> row sharing a <code>productGroupId</code> with siblings
Tombstone	<code>deletedAt</code> non-null soft-delete marker on <code>products</code>
Snapshot fields	VAT/discount/shipping captured on the order at checkout
Loyalty AED	Wallet balance redeemable as a discount in AED